EAST WEST UNIVERSITY

Title

Project Report

Course Code: CSE 207

Course Title: Data Structure and Algorithm

Semester: Spring 2026

Section: 10

Course Instructor

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Bank Management System Project Report

Course Information

Course Title: Data Structures (CSE207)
Institution: East West University
Semester: Spring 2025

Project Title

Bank Management System Using C Programming

Objective

The objective of this project is to develop a simple Bank Management System using C programming. It simulates real banking operations and demonstrates the use of data structures like Linked List and Stack.

Features

1. Login System

```c
/* — A. LOGIN — */
int loginInterface() {
    char user[30], pass[10];
    clearScreen();
    printHeader("LOGIN");
    printf("\n  Username (your nickname)        : ");
    scanf("%29s", user);
    printf("  Password (last 3 digits of ID)  : ");
    scanf("%9s",  pass);

    Account *found = findAccount(user);
    if (found && strcmp(found->password, pass) == 0) {
        currentUser = found;
        printf("\n  >> Login successful! Welcome, %s.\n", user);
        pressEnter();
        return 1;
    }
    printf("\n  >> Invalid username or password!\n");
    pressEnter();
    return 0;
}
```

2. Balance Check

```c
void checkBalance() {
    clearScreen();
    printHeader("BALANCE CHECK");
    printf("\n  Account Holder    : %s\n",      currentUser->username);
    printf("  Current Balance   : %.2f TK\n", currentUser->balance);
    printf("\n  (All accounts start with %.2f TK)\n", INITIAL_BAL);
    pressEnter();
}
```

3. Account Statement

```c
void printStatement() {
    clearScreen();
    printHeader("ACCOUNT STATEMENT");

    Stack *s = &currentUser->history;
    if (stackEmpty(s)) {
        printf("\n  No transactions found.\n");
        pressEnter();
        return;
    }

    printf("\n  %-4s  %-30s  %-4s  %-12s  %-12s\n",
           "No.", "Description", "Type", "Amount(TK)", "Balance(TK)");
    printLine('-', 68);

    for (int i = 0; i <= s->top; i++) {
        printf("  %-4d  %-30s  %-4s  %-12.2f  %-12.2f\n",
                i + 1,
                s->data[i].desc,
                s->data[i].type,
                s->data[i].amount,
                s->data[i].balAfter);
    }
    printLine('-', 68);
    printf("  Total transactions : %d\n", s->top + 1);
    printf("  Current Balance    : %.2f TK\n", currentUser->balance);
    pressEnter();
}
```

4. Deposit System

```
void depositMoney() {
    float amount;
    clearScreen();
    printHeader("DEPOSIT");
    printf("\n  Current Balance  : %.2f TK\n", currentUser->balance);
    printf("  Deposit amount   : ");
    scanf("%f", &amount);

    if (amount <= 0) { printf("\n  >> Invalid amount!\n"); pressEnter(); return; }

    currentUser->balance += amount;
    recordTxn(currentUser, "Cash Deposit", amount, "CR");
    printf("\n  >> Deposit successful!\n");
    printf("  New Balance : %.2f TK\n", currentUser->balance);
    pressEnter();
}
```

5. Withdrawal System

```
void withdrawMoney() {
    float amount;
    clearScreen();
    printHeader("WITHDRAW");
    printf("\n  Current Balance  : %.2f TK\n", currentUser->balance);
    printf("  Withdraw amount  : ");
    scanf("%f", &amount);

    if (amount <= 0)                    { printf("\n  >> Invalid amount!\n");       pressEnter(); return; }
    if (amount > currentUser->balance) { printf("\n  >> Insufficient balance!\n"); pressEnter(); return; }
                                  float Account::balance
    currentUser->balance -= amount;
    recordTxn(currentUser, "Cash Withdrawal", amount, "DR");
    printf("\n  >> Withdrawal successful!\n");
    printf("  New Balance : %.2f TK\n", currentUser->balance);
    pressEnter();
}
```

6. Online Payment System

```
void paymentSystem() {
    char shopName[40], shopAcc[30];
    float amount;
    clearScreen();
    printHeader("PAYMENT - ONLINE SHOPPING");

    printf("\n  Current Balance         : %.2f TK\n", currentUser->balance);
    printf("\n  Enter shop name         : "); scanf("%39s", shopName);
    printf("  Enter shop account no.  : "); scanf("%29s", shopAcc);
    printf("  Enter payment amount    : "); scanf("%f",   &amount);

    if (amount <= 0)                    { printf("\n  >> Invalid amount!\n");       pressEnter(); return; }
    if (amount > currentUser->balance) { printf("\n  >> Insufficient balance!\n"); pressEnter(); return; }

    currentUser->balance -= amount;
    char desc[60];
    snprintf(desc, 59, "Payment->%.15s[%.10s]", shopName, shopAcc);
    recordTxn(currentUser, desc, amount, "DR");

    printf("\n  -------- PAYMENT RECEIPT --------\n");
    printf("  Shop          : %s\n",       shopName);
    printf("  Account No.   : %s\n",       shopAcc);
    printf("  Amount Paid   : %.2f TK\n",  amount);
    printf("  New Balance   : %.2f TK\n",  currentUser->balance);
    printf("  Status        : SUCCESS\n");
    pressEnter();
}
```

7. Loan System

```c
void loanSystem() {
    char  loanNames[3][3][30] = {
        {"House Loan",      "Car Loan",        "Education Loan"},
        {"Car Loan",        "Education Loan",  ""},
        {"Education Loan",  "",                ""}
    };
    float loanMax[3][3] = {
        {1000000.0f, 500000.0f, 200000.0f},
        { 300000.0f, 150000.0f,       0.0f},
        {  50000.0f,      0.0f,       0.0f}
    };
    int loanCount[3] = {3, 2, 1};

    float income;
    clearScreen();
    printHeader("LOAN SYSTEM");
    printf("\nEnter your monthly income: ");
    scanf("%f", &income);

    int tier;
    if      (income >= 50000) tier = 0;
    else if (income >= 20000) tier = 1;
    else                      tier = 2;

    printf("\n====== LOAN OPTIONS ======\n");
    for (int i = 0; i < loanCount[tier]; i++)
        printf("  %d. %-22s  Up to: %.0f TK\n", i+1, loanNames[tier][i], loanMax[tier][i]);
    printf("==========================\n");

    int ch;
    printf("\n  Select loan option (0 = cancel) : ");
    scanf("%d", &ch);
    if (ch < 1 || ch > loanCount[tier]) { printf("  >> Cancelled.\n"); pressEnter(); return; }

    float req;
    printf("  Enter loan amount requested     : ");
    scanf("%f", &req);

    if (req <= 0 || req > loanMax[tier][ch-1]) {
        printf("\n  >> Amount invalid or exceeds maximum (%.0f TK)!\n", loanMax[tier][ch-1]);
        pressEnter(); return;
    }

    currentUser->balance += req;
    char desc[60];
    snprintf(desc, 59, "Loan: %s", loanNames[tier][ch-1]);
    recordTxn(currentUser, desc, req, "CR");

    printf("\n  >> Loan of %.2f TK approved and credited!\n", req);
    printf("  New Balance : %.2f TK\n", currentUser->balance);
    pressEnter();
}
```

8. Interest System

```c
void interestSystem() {
    clearScreen();
    printHeader("INTEREST SYSTEM");

    float bal  = currentUser->balance;
    float rate;

    if      (bal >= 100000) rate = 8.5f;
    else if (bal >=  50000) rate = 7.0f;
    else if (bal >=  20000) rate = 5.5f;
    else if (bal >=  10000) rate = 4.0f;
    else                    rate = 2.5f;

    printf("\n  Current Balance       : %.2f TK\n",   bal);
    printf("  Your Interest Rate    : %.1f%% p.a.\n", rate);
    printf("\n  -- Balance Range & Interest Rates --\n");
    printf("  100,000+ TK  -->  8.5%%\n");
    printf("   50,000+ TK  -->  7.0%%\n");
    printf("   20,000+ TK  -->  5.5%%\n");
    printf("   10,000+ TK  -->  4.0%%\n");
    printf("    < 10,000   -->  2.5%%\n");

    int years;
    printf("\n  Enter number of years : ");
    scanf("%d", &years);
    if (years <= 0) { printf("  >> Invalid!\n"); pressEnter(); return; }

    float future = compoundInterest(bal, rate, years);
    float earned = future - bal;

    printf("\n  -------- INTEREST SUMMARY --------\n");
    printf("  Principal        : %.2f TK\n", bal);
    printf("  Rate             : %.1f%% per year\n", rate);
    printf("  Years            : %d\n",      years);
    printf("  Interest Earned  : %.2f TK\n", earned);
    printf("  Total Amount     : %.2f TK\n", future);
    printf("  (Computed using recursive compound interest)\n");
    pressEnter();
}
```

Data Structures Used

Linked List: Used to store user accounts.
Stack: Used to store transaction history.

Conclusion

This project demonstrates how banking operations can be implemented using C programming and data structures. It helps in understanding real-life applications of programming.